

INTERACTIVE AND DIGITAL TOOLS

Grado en Diseño/Bachelor in Design BID SEP-2023 IDT-D.2.S.A

Area Architecture and Design

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Semester: 1º

Category: COMPULSORY

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María is a researcher and Professor of Architecture and Design whose work focuses on the use of new platforms and techniques for digital innovation with an emphasis on AI and its ethical implications in the design of new technologies.

María graduated from Massachusetts Institute of Technology (MIT) with a Master of Science in Architecture Studies. Her individual research on machine learning, ethics, and design was recognized with the MIT Thesis Prize. During her time at MIT, she taught undergraduate and graduate design studios in the disciplines of Architecture and Design.

María is a former researcher at MIT's Self-Assembly Lab, where she worked on self-assembly and programmable material technologies for novel manufacturing, products, and construction processes. She also became a collaborator at Harvard University, developing new bioinspired materials at the Wyss Institute for Biologically Inspired Engineering.

María has designed and built installations and exhibitions at galleries around the world (Boston, Chicago, Edinburgh, LA, London, Madrid, Rome, Seoul, and Venice) including the I Seoul Biennale of Architecture and Urbanism, and Venice Architecture Biennales. Her work has been featured in international publications, for which she has received various awards and scholarships. María has lectured at different institutions including Korea University, University of Plymouth, and the Architectural Association Visiting School Madrid.

In 2017, María was selected as the Rome Prize in Architecture by the Spanish Academy in Rome.

In 2019 María was named one of the "Top 24 Most Innovative and Radical Female Artists in Spain" by El Mundo magazine, and in 2020, she was included in the list of "Spaniards who are Changing the World" from La Razon newspaper.

Prior to MIT, she received her Master's Degree in Architecture from the Bartlett School of Architecture (UCL) and a Master of Arts in Architecture from the University of Edinburgh.

Previously, María has worked at several renowned design and architecture offices including Grimshaw Architects (London) and Dominique Perrault (Paris).

Right now, María is leading design and research projects at MITdesignX, an academic program in the MIT School of Architecture and Planning dedicated to design innovation and entrepreneurship. She is also Adjunct Professor at IE University School of Architecture and Design in Madrid, receiving the 2021-22 Award for Teaching Excellence.

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I will hold virtual office hours over zoom outside of the listed class time. You are encouraged to come to my office hours, which will be held by appointment from Monday to Friday. Although office hours are optional, please do communicate with me if you feel like you are struggling or feeling lost at any point in the process. My door is always open!

SUBJECT DESCRIPTION

Welcome to Interactive + Digital Tools, a course that will introduce you to digital tools and emerging technologies relevant to contemporary design practice. In this course you will learn how to use software in your design process and the role that technology plays in creativity and visual communication. This course is structured in three units with specific deliverables at the end of each unit. The course will consist of project-based explorations that will challenge you to re-think the way you approach software. You will be asked to navigate the space between technology and design.

Given the context of the IE School of Architecture & Design which places emphasis on excellence in design and innovation, the relation between these two concepts will be foregrounded continually. The course aims to offer critical acumen in respect of interactive and digital tools through creative design work. A sequence of lectures, workshops, ideation sessions, desk crits, and studio reviews will provide a framework for you to develop your projects. This classroom will be considered a laboratory where experimentation is encouraged.

In order to prepare for each class, weekly course assignments are to be completed and posted to MIRO prior to our class meetings. This will help you engage more in the class discussions, as well as understand your classmates' take on each assignment. As an instructor, it will give me the chance to review your work prior to our class.

Even though you are not required to literally carry all the lessons and findings from one exercise to the other, the sequence is designed to progressively engage with topics, references, and design strategies. The assignments are designed to help you propel your project forward while exploiting the collective intelligence of the entire class. That is why active participation is essential in the classroom, accounting for 20% of the final grade.

Studio Culture: Work in the design studio will build sequentially. Therefore, your commitment to continual development on a daily basis is of paramount importance. I want to emphasize the importance of your peers as a source of support, inspiration, and feedback. Magnification of your development as a designer is made possible by the collective nature of the class. Group reviews are collective for a reason. Our studio is a place for all, and it requires the careful attention to the needs of everyone in it.

Participation: Significant deductions will be made for students who don't engage or participate. Laptops are essential tools in this class, but should not be used during lectures. We have limited class time for each project, so make use of it! Starting late or leaving early is not an option—even if it's a working session.**Inclusivity Statement:** I hope to foster a sense of community in this classroom and consider this classroom to be a place where you will be treated with respect. I welcome individuals of all backgrounds, beliefs, ethnicities, national origins, gender identities, sexual orientations, religious and political affiliations – and other visible and nonvisible differences. All members of this class are expected to contribute to a respectful, welcoming, and inclusive environment for every other member of the class. If this standard is not being upheld, please feel free to speak with me.

LEARNING OBJECTIVES

The course is an introduction to digital tools and emerging technologies relevant to contemporary design practice. By the end of this course, you will be able to:

- Move fluidly between different digital tools, modes, and scales of operation.
- Know how to combine traditional manual techniques with digital media.
- Use design skills and digital tools to turn your ideas into developed design proposals.
- Integrate software in the wider workflow of a project.
- Use technology for creativity and visual communication.
- Engage with an increasing level of design-research.
- Work in an iterative process (feedback - loop process).

Through lectures and weekly exercises, students will develop their skills and enable creativity, abstract thinking, representation, iteration, and design development. Throughout the course, you will acquire the appropriate level creative expression and develop individual criteria regarding the use and combination of several digital tools during conceptual design, production, and communication processes. You will also develop your own critical analysis skills, meaning that by the end of the course, you should be able to critically evaluate your own design work, as well as the methodologies and tools that inform them. Conventions of design representation and communication will be explored. You will need to demonstrate an understanding of conventions, and an ability to sustain an increasing level of research in the projects over the semester. You will develop your ability to conceptually develop, produce, and communicate design projects by means of 2-, 2.5-, and 3-dimensional digital tools, and be able to translate information into interaction. Finally, you will gain skills on how to use digital tools in a creative manner.

TEACHING METHODOLOGY

IE University teaching method is defined by its collaborative, active, and applied nature. Students actively participate in the whole process to build their knowledge and sharpen their skills. Professor's main role is to lead and guide students to achieve the learning objectives of the course. This is done by engaging in a diverse range of teaching techniques and different types of learning activities such as the following:

This course is centered around project-based learning assignments, where you will investigate and actively respond to different design challenges. Projects are expected to be developed during the class sessions and completed afterwards, and will span multiple sessions, allowing you to put into practice the newly acquired concepts. Many of this course's sessions will be focused around desk crits and reviews, allowing you to work in an iterative process (feedback - loop), an essential skill in the design world. Such sessions also allow for a critical dialogue to emerge, involving the active participation of the entire class.

Outside of our class sessions, you will be guided to independently learn the technical aspects through the infinite potential of online learning resources. In between class sessions, you will be able to get further feedback on your work via office hours.

To prepare for each class, course assignments are to be completed and posted on MIRO prior to our class meetings. MIRO will be used as the main online platform for this course as it will provide you with an easy access to all the required learning resources as well as a space where you can upload your work, share references, and give/receive feedback. I will be also posting my office hours on MIRO for those who would like extra input. At the beginning of the course, I will share and upload to MIRO a detailed schedule, including the due dates for all the assignments.

Learning Activity	Weighting	Estimated time a student should dedicate to prepare for and participate in
Lectures	10.0 %	15.0 hours
Discussions	10.0 %	15.0 hours
Exercises in class, Asynchronous sessions, Field Work	10.0 %	15.0 hours
Group work	13.33 %	20.0 hours
Individual studying	56.67 %	85.0 hours
TOTAL	100.0 %	150.0 hours

PROGRAM

PROJECT 1: THE I+D TOOLS DECK

The I+D Design Deck is a deck of playing cards that doubles as a practical guide and introduction to design, its tools, and technologies. With this first exercise, you will learn the basics of visual composition and develop a creative proposal linked to a specific concept. The objective of this exercise is for you to turning a given concept into a graphic component that unifies and transmits your take on the topic. By the end of the project, you will be able to visually communicate complex messaging by carefully synthesizing and structuring information.

Sessions 1 - 7.

Presentation + Deliverables: 20% of grade.

SESSIONS 1 - 2 (LIVE IN-PERSON)

Class Introduction / Course goals and objectives / Course structure.

Project 1 introduction.

SESSIONS 3 - 4 (LIVE IN-PERSON)

Student presentations followed by work session with desk crits.

SESSIONS 5 - 7 (LIVE IN-PERSON)

Project 1 – Final Review.

PROJECT 2: TIME-BASED MEDIA

We will introduce different workflows for creating time-based media. After dissecting a fragment of the film, create a piece of time-based media that projects the original film clip into the future. Create a concept and narrative that take a stance on the original and reinterpret it and convey it in this new moving image.

Sessions 8 - 19.

Presentation + Deliverables: 30% of grade.

SESSIONS 8 - 9 (LIVE IN-PERSON)

Together with your partner, select 4 short clips (15sec aprox. each). Add all four clips to your MIRO boards explaining why you picked those scenes. Think about the architectural elements when picking those scenes, and comment on the cinematic techniques (for example, you may comment on the framing, angle, and camera movement of a shot, sound and editing used in that scene, as well as costumes, props, sets, etc.).

Make a diagram about that particular scene. For example, you might chose to represent the movements of the characters.

SESSIONS 10 - 11 (LIVE IN-PERSON)

Decide on a narrative, pick the view or views you'd like to render, and decide what will change over the course of your new clip. Sketch a storyboard. Think about time of day, environment, material weathering, construction, program, scale figures. Make a diagram that shows the overall intention of this piece of time-based media.

SESSIONS 12 - 14 (LIVE IN-PERSON)

Desk crits.

Individual meetings for feedback on your progress so far.

SESSIONS 15 - 16 (LIVE IN-PERSON)

Export the segments of the film that you have so far, even if they are work in progress. Upload them to MIRO. Give feedback to a minimum of two teams on their time-based media segments.

SESSIONS 17 - 19 (LIVE IN-PERSON)

Project 2 - Final Review.

PROJECT 3: APP

You will be asked to design an app. You will build, experiment and test your process throughout the rest of the semester. At the end of this project, you will develop the concept and narrative about your app, how it fits into a context, how it works and explore proposals for future applications.

Sessions 20 - 30

Presentation + Deliverables: 30% of grade.

SESSIONS 20 - 21 (LIVE IN-PERSON)

Project 3 introduction.

Group formation exercise followed by a work session.

SESSIONS 22 - 23 (LIVE IN-PERSON)

Desk crits followed by work session.

SESSIONS 24 - 25 (LIVE IN-PERSON)

Informal student presentations.

SESSIONS 26 - 27 (LIVE IN-PERSON)

Desk Crits.

Last class meeting before final review.

SESSIONS 28 - 30 (LIVE IN-PERSON)

Final Review.

EVALUATION CRITERIA

At the beginning of each project, you will be given a detailed project brief accompanied by a grading rubric with explicit set of criteria that will be used to assess your work.

The grading rubric will be specific to each assignment, and will include marking criteria, levels of performance, scores, and descriptors.

Class discussions and participation includes an active participation during class session.

If you are caught committing plagiarism, you will automatically fail the assignment that you plagiarized. Plagiarism on a second assignment is an automatic failure of this class.

You are required to submit all material to receive credit and a grade for the course.

If at any point you are unsure of how you are being graded, please let me know.

Evaluation is continuous, which implies that all the work produced by the students along the semester will contribute to the final grade. Final grade for the course is calculated as the weighted average in the following percentages:

criteria	percentage	Learning Objectives	Comments
Project 1	20 %		
Project 2	30 %		
Project 3	30 %		
Class Discussions, Participation, Attendance	20 %		

As per IE University policy, students who do not comply with the 70% attendance rule will automatically fail and will be required to enrol again in this course next academic year.

Grading for retakes will be subject to the following rules:

- Students failing the course in the first ordinary period will have to do a retake in June (except those not complying with the attendance rules, which are banned from this possibility).
- Dates and location of the June retakes will be posted a month in advance and will not be changed. Please take this into consideration when planning your summer.
- The June retakes will consist in an individual project. The grade will depend only on the performance in this project. The passing grade is 5 out of 10. The maximum grade that a student may obtain in any type of retake will be 8 out of 10.
- Students who fail to obtain a satisfactory grade in the 1st or 2nd Periods will be required to enrol again in this course next academic year.

BEHAVIOR RULES

Please, check the University's Code of Conduct [here](#). The Program Director may provide further indications.

ATTENDANCE POLICY

Please, check the University's Attendance Policy [here](#). The Program Director may provide further indications.

According to the current attendance policy of IE University, class attendance is mandatory: students must attend at least 70% of all class sessions. The studio is a great learning environment that requires your physical presence as well as your intellectual presence. Students' attendance is not determined by mere passive presence, but by an active participation in class: students must participate in class activities to be accounted for as attending the class.

Absences of any kind are strongly discouraged as your learning and work will be adversely affected by the information and activities you miss. Be punctual, arriving just before the class start time so we can begin sessions promptly, and stay for the duration of each class. If you are five minutes late or leave class early you will be marked as absent.

ETHICAL POLICY

Please, check the University's Ethics Code [here](#). The Program Director may provide further indications.

In this course, I will hold you to the high standard of academic integrity expected of all students at IE University. I do this for two reasons. First, it is essential to the learning process that you are the one doing the work. I have structured the assignments in this course to enable you to gain a mastery of the course material. Failing to do the work yourself will result in a lesser understanding of the content, and therefore a less meaningful education for you. Second, it is important that there be a level playing field for all students in this course and at the Institute so that the rigor and integrity of the Institute's educational program is maintained.

All the work that a student submits has to be original and produced by the student: not copied nor downloaded from the internet, unless students are specifically instructed to copy other's work as a part of the actual exercise.

Please review the [IE Code of Ethical Conduct](#), where you will be able to read the IE's Academic Standards. Contact me if you have any questions about appropriate citation methods, the degree of collaboration that is permitted, or anything else related to the Academic Integrity of this course.

EMAIL POLICY

Please reach out to me if you have any questions or if you want to meet outside of my office hours. However, you should not rely on an immediate response to questions sent via email/MIRO. Please begin assignments with enough time to attend office hours or ask a question several days before the assignment is due.

PLEASE TAKE CARE OF YOURSELF

If you feel overwhelmed with the demands of this course, if you feel like you might miss a deadline or that you might need to copy work in order to get things done, contact me right away and we will work out a plan. It is better than plagiarizing or not coming to class.

MENTAL HEALTH

As a student, you may experience a range of challenges that can interfere with learning, such as strained relationships, increased anxiety, feeling down, difficulty concentrating and/or lack of motivation. These mental health concerns or stressful events may impact your ability to attend class, concentrate, complete work, take an exam, or participate in daily activities.

The student community at IEU has access to their counselling service during the academic year. This service is free and confidential. The team is comprised of guidance counsellors (IEU staff) and clinical psychologists and psychiatrists from SINEWS, a mental health service provider.

To schedule an appointment:

- Email: Counseling.IEU@ie.edu (or Mentoring.IEU@ie.edu).
- Phone: (+34) 921 415 312
- 24/7 Mental Health Crisis Line: (+34) 619 270 148 (English/Spanish).

For urgent or after-hours concerns, please dial 112 immediately.